

ONLINE APPENDIX: DISCRETE CHOICE WITH GENERALIZED SOCIAL INTERACTIONS

OSCAR VOLPE

Department of Economics, Harvard University

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This online appendix accompanies the paper “Discrete Choice with Generalized Social Interactions”. It contains supplementary proofs, model extensions, additional details about identification and estimation, and robustness analyses for the empirical applications. All notation is consistent with the main text of the paper.

APPENDIX A: EQUILIBRIUM ANALYSIS AND MODEL EXTENSIONS

A.1. Proofs of Supplementary Results

LEMMA A.1: *If $J_{kj_1}J_{j_1j_2}\cdots J_{j_Mk} = 0$ for every integer $M \geq 1$ and all types $k, j_1, \dots, j_M$, then $\rho(\mathbf{D}(m)) = 0$ for all $m \in [0, 1]^K$. Hence, there is a unique, locally stable equilibrium.*

PROOF: Taking $M = 1$ and $j_1 = k$ gives $J_{kk}^2 = 0$, so $J_{kk} = 0$. It follows that $(\mathbf{J}^M)_{kk} = 0$ for all types k and integers $M \geq 1$. Since this property also applies to $\mathbf{D}(m)$, it must be that:

$$\sum_{k=1}^K [\lambda_k(m)]^M = \text{tr}([\mathbf{D}(m)]^M) = \sum_{k=1}^K ([\mathbf{D}(m)]^M)_{kk} = 0,$$

for every $M \geq 1$, where $\{\lambda_k(m)\}_k$ denote the eigenvalues of $\mathbf{D}(m)$, counted with algebraic multiplicity. By Newton’s identities, the vanishing of the first K power sums guarantees that $\lambda_k(m) = 0$ for all k . Given that $\rho(\mathbf{D}(m)) = \max_k \{|\lambda_k(m)|\}$, it follows that $\rho(\mathbf{D}(m)) = 0$. The index theorem implies the equilibrium is unique. Local stability holds from Lemma 1.

Q.E.D.

LEMMA A.2: *There exist positive thresholds S_k , which depend only on $f_{\varepsilon|k}$ and h_k , so that, whenever $\sum_{\ell=1}^K |J_{k\ell}| < S_k$ for every type k , the equilibrium is unique and locally stable.*

PROOF: By [Horn and Johnson \(1985\)](#), Thm. 5.6.9, it follows that $\rho(\mathbf{D}(m)) \leq \|\mathbf{D}(m)\|_\infty$, where $\|\cdot\|_\infty$ represents the p -norm defined by the maximum absolute row sum of $\mathbf{D}(m)$:

$$\|\mathbf{D}(m)\|_\infty = \max_k \left\{ \sum_{\ell=1}^K |[\mathbf{D}(m)]_{k\ell}| \right\} = \max_k \left\{ f_{\varepsilon|k} \left(h_k + \sum_{\ell=1}^K J_{k\ell} m^\ell \right) \sum_{\ell=1}^K |J_{k\ell}| \right\}.$$

For any type k , choose $S_k > 0$ such that $S_k \max_{|z-h_k| \leq S_k} f_{\varepsilon|k}(z) = 1$. Such a threshold exists because the left-hand side is continuous in S_k , tends to zero as $S_k \rightarrow 0$, and tends to infinity as $S_k \rightarrow \infty$. Its construction depends only on h_k and $f_{\varepsilon|k}$. If $\sum_{\ell=1}^K |J_{k\ell}| < S_k$, then for all m :

$$\left| \sum_{\ell=1}^K J_{k\ell} m^\ell \right| \leq \sum_{\ell=1}^K |J_{k\ell}| < S_k.$$

Therefore, $f_{\varepsilon|k}(h_k + \sum_{\ell=1}^K J_{k\ell} m^\ell) \leq 1/S_k$. If the condition holds for every type k , then:

$$\sup_{m \in [0,1]^K} \|\mathbf{D}(m)\|_\infty \leq \max_k \left\{ \frac{1}{S_k} \sum_{\ell=1}^K |J_{k\ell}| \right\} < 1.$$

Thus, the equilibrium mapping is a contraction, with a unique and locally stable fixed point.
Q.E.D.

LEMMA A.3: *If $\mathbf{J}$ is symmetric and its eigenvalues all have nonpositive real parts, then there is a unique equilibrium, locally stable if $\rho(\mathbf{D}(m)) < 1$ and unstable if $\rho(\mathbf{D}(m)) > 1$.*

PROOF: Let $\mathbf{J}$ be symmetric with eigenvalues that all have nonpositive real parts. Since $\beta(m)$ has nonnegative diagonal entries, $\beta^{1/2}(m)\mathbf{J}\beta^{1/2}(m)$ is symmetric negative semidefinite. Additionally, this matrix and $\mathbf{D}(m) = \beta(m)\mathbf{J}$ both share the same characteristic polynomial, because AB and BA have the same characteristic polynomial, taking $A = \beta^{1/2}(m)$ and $B = \beta^{1/2}(m)\mathbf{J}$. Therefore, all eigenvalues of $\mathbf{D}(m)$ are real and nonpositive. It follows that: $0 < \prod_{k=1}^K (1 - \lambda_k(m)) = \det(I - \mathbf{D}(m)) = \det(\mathbf{D}_{\mathcal{H}}(m))$ at any equilibrium m . By the index theorem, the model always has a unique equilibrium. By Lemma 1, this equilibrium will be locally stable whenever $\rho(\mathbf{D}(m)) < 1$, and it will be unstable whenever $\rho(\mathbf{D}(m)) > 1$.
Q.E.D.

LEMMA A.4: *Suppose that the interaction matrix $\mathbf{J}$ is irreducible. Then the No Negative Feedback condition holds if and only if $\mathbf{B}\mathbf{D}(\cdot)\mathbf{B}^{-1}$ is nonnegative for a diagonal matrix $\mathbf{B}$.*

PROOF: Because $\mathbf{B}$ and $\beta(m)$ are diagonal, it follows that $\mathbf{B}\mathbf{D}(m)\mathbf{B}^{-1} = \beta(m)\mathbf{B}\mathbf{J}\mathbf{B}^{-1}$. Hence, nonnegativity of $\mathbf{B}\mathbf{J}\mathbf{B}^{-1}$ is equivalent to nonnegativity of $\mathbf{B}\mathbf{D}(m)\mathbf{B}^{-1}$ at every m .

“ $\Leftarrow$ ” To begin, suppose that $\mathbf{J}$ is similar to a nonnegative matrix $\mathbf{A}$ by way of a diagonal change-of-basis matrix $\mathbf{B}$. Specifically, there exists a diagonal $\mathbf{B}$ where $\mathbf{A} = \mathbf{B}\mathbf{J}\mathbf{B}^{-1} \geq \mathbf{0}$. Since $\mathbf{B}$ is diagonal, the elements of $\mathbf{A}$ can be expressed as $A_{k\ell} = B_{kk}J_{k\ell}/B_{\ell\ell}$, for all k, ℓ . Therefore, for any choice of indices k and $\ell_1, \dots, \ell_M$ in the set $\{1, \dots, K\}$, it must be that:

$$\begin{aligned} 0 \leq A_{k\ell_1} A_{\ell_1\ell_2} \cdots A_{\ell_M k} &= \left(\frac{B_{kk}}{B_{\ell_1\ell_1}} J_{k\ell_1} \right) \left(\frac{B_{\ell_1\ell_1}}{B_{\ell_2\ell_2}} J_{\ell_1\ell_2} \right) \cdots \left(\frac{B_{\ell_M\ell_M}}{B_{kk}} J_{\ell_M k} \right) \\ &= J_{k\ell_1} J_{\ell_1\ell_2} \cdots J_{\ell_M k}. \end{aligned}$$

“ $\Rightarrow$ ” Suppose that NNF holds. When $\mathbf{J}$ is irreducible, all nonzero products of interaction effects along paths from k to ℓ share the same sign; otherwise, appending the same return path with a nonzero product would make one feedback effect negative. For every k , define $\{\gamma_\ell^k\}_{\ell=1}^K$ so that $\gamma_\ell^k = 1$ if $J_{kj_1} J_{j_1 j_2} \dots J_{j_M \ell} \geq 0$ for all integers $M \geq 1$ and types $j_1, \dots, j_M$, and $\gamma_\ell^k = -1$ otherwise. Next, fixing some reference type k_0 , define the matrix $\mathbf{B}$ such that:

$$\mathbf{B} = \text{diag} \begin{bmatrix} \gamma_1^{k_0} \\ \vdots \\ \gamma_K^{k_0} \end{bmatrix}.$$

Notice that $\mathbf{B}$ is involutory, i.e. $\mathbf{B}^{-1} = \mathbf{B}$. Therefore, for all (k, ℓ) , NNF would ensure that:

$$[\mathbf{B}\mathbf{J}\mathbf{B}^{-1}]_{k,\ell} = [\mathbf{B}\mathbf{J}\mathbf{B}]_{k,\ell} = \gamma_k^{k_0} \gamma_\ell^{k_0} J_{k\ell} = \gamma_{k_0}^k \gamma_\ell^{k_0} J_{k\ell} = \gamma_\ell^k J_{k\ell} = |J_{k\ell}|$$

It follows that $\mathbf{B}\mathbf{J}\mathbf{B}^{-1}$ equals the absolute value of $\mathbf{J}$. Therefore, $\mathbf{B}\mathbf{J}\mathbf{B}^{-1}$ is nonnegative. *Q.E.D.*

LEMMA A.5: *No Negative Feedback holds if $\text{sgn}(J_{k\ell}) = \text{sgn}(J_{km}J_{m\ell})$ for all k, ℓ , and m .*

PROOF: To begin, suppose that $\text{sgn}(J_{k\ell}) = \text{sgn}(J_{km}J_{m\ell})$ for any types k, ℓ, m . I prove by induction that $\text{sgn}(J_{j_0 j_1} J_{j_1 j_2} \dots J_{j_M j_0}) \geq 0$ for any arbitrary types $j_0, j_1, \dots, j_M$. First, note that $\text{sgn}(J_{j_0 j_2}) = \text{sgn}(J_{j_0 j_1} J_{j_1 j_2})$. Next, let $\text{sgn}(J_{j_0 j_M}) = \text{sgn}(J_{j_0 j_1} J_{j_1 j_2} \dots J_{j_{M-1} j_M})$ for some positive integer $M \geq 1$. By construction, the following equation must be satisfied:

$$\text{sgn}(J_{j_0 j_{M+1}}) = \text{sgn}(J_{j_0 j_M} J_{j_M j_{M+1}}) = \text{sgn}(J_{j_0 j_1} J_{j_1 j_2} \dots J_{j_{M-1} j_M} J_{j_M j_{M+1}}).$$

By induction, $\text{sgn}(J_{j_0 \ell}) = \text{sgn}(J_{j_0 j_1} J_{j_1 j_2} \dots J_{j_M \ell})$ for any agent type ℓ . By setting $\ell = j_0$, I obtain that $\text{sgn}(J_{j_0 j_1} J_{j_1 j_2} \dots J_{j_M j_0}) = \text{sgn}(J_{j_0 j_0})$ where $\text{sgn}(J_{j_0 j_0}) = \text{sgn}(J_{j_0 j_0} J_{j_0 j_0}) \geq 0$. *Q.E.D.*

LEMMA A.6: *Assumption A implies that, for each integer $r \geq 1$ and vector $m \in [0, 1]^K$:*

$$\sum_{(k_0, \dots, k_{r-1}) \in \{1, \dots, K\}^r} \left[\prod_{q=0}^{r-1} \beta_{k_q}(m) \right] J_{k_0 k_1} J_{k_1 k_2} \dots J_{k_{r-1} k_0} \geq 0,$$

where $\beta_k(m) = [\beta(m)]_{kk}$ and, in the case where $r = 1$, the interaction product equals $J_{k_0 k_0}$.

PROOF: Assumption A implies $\mathbf{A}(m) = \mathbf{B}\mathbf{D}(m)\mathbf{B}^{-1}$ is entrywise nonnegative, so $\mathbf{A}(m)^r$ is entrywise nonnegative for every $r \geq 1$. As trace is invariant under similarity, $\text{tr}(\mathbf{D}(m)^r) = \text{tr}(\mathbf{A}(m)^r) \geq 0$. Expanding $\text{tr}(\mathbf{D}(m)^r)$ and substituting $[\mathbf{D}(m)]_{k\ell} = \beta_k(m) J_{k\ell}$ gives the result. *Q.E.D.*

A.2. Extensions to Alternative Network-Based Models

The properties laid out in Section 2 are not specific to binary choice models. They also have implications for a much broader class of models where agents interact in a network. To see how, consider a game with K players. Each player k chooses a_k from a compact action space $A_k \subset \mathbb{R}$. Given a profile of actions $a \in A_1 \times A_2 \times \cdots \times A_K$, each player's best response is $a_k^* = q_k(\sum_{\ell=1}^K J_{k\ell} a_\ell)$, where $q_k(\cdot)$ is a non-decreasing mapping from $\mathbb{R}$ to A_k . The game permits, but does not require, each player's response to directly depend on her own action.

This game encompasses a variety of economic models. For example, the action a_k could represent a person's investment into a public good, where everyone benefits from how much their neighbors contribute.¹ Alternatively, a_k could represent the output produced by a firm that competes in an oligopoly, where every firm's action affects the market price. Both these models are well-studied, and they both involve negative interaction effects between agents.

Another interpretation of the game is that each player is a community of individuals. In the binary choice model, a player is a social group whose members make one of two choices subject to social influences. Agents act noncooperatively, and a_k is the average choice in group k . The same logic applies when residents of a state or local institution k make a collective choice a_k . For example, one could model policy spillovers across U.S. states, where voters support more liberal or conservative positions in response to laws adopted elsewhere.

I focus on pure strategy Nash equilibria, defined as the action profiles a from which no player k wishes to deviate. An equilibrium will exist if there are continuous best responses and compact, convex action spaces. Moreover, if the best responses are continuously differentiable, then the spectral radius of the Jacobian matrix provides a sufficient statistic for determining uniqueness. To understand how, let a^* be the equilibrium at which $\rho(\mathbf{D}(a))$ is greatest. If $\rho(\mathbf{D}(a^*)) < 1$, then a^* is a unique, locally stable equilibrium. Conversely, under Assumption A, if $\rho(\mathbf{D}(a^*)) > 1$, then there are generically multiple, locally stable equilibria.

The set of equilibria is generally not well-ordered. In particular, it need not form a lattice structure, which would guarantee greatest and least equilibria at which every player takes her highest and lowest equilibrium actions, respectively. Property 4 illustrates how negative interactions can place different players' highest equilibrium actions at different equilibria, and it considers the implications for social welfare that depend on the underlying payoffs.

¹Bramoullé et al. (2014) study a public goods game that is nested within this framework. In their paper, players choose from an interval $[0, 1]$ and the best responses are $a_k = \max\{0, 1 - \delta \sum_{\ell \neq k} g_{k\ell} a_\ell\}$, where $\delta \in [0, 1]$ and $g_{k\ell} \in \{0, 1\}$ indicates whether players k and ℓ are linked. Note that this game involves pure strategic substitutes.

A.3. Endogenous Network Formation with Generalized Interactions

In the paper, it is assumed that agents respond to the expected average action for each type. I now consider an alternative model, where utility depends on the expected composition of agents who choose an action; that is, agents care about $P(k = a|\omega_i)$ instead of $E(\omega_i|k = a)$.

This alternative model specification can be used to analyze how social identity influences network formation. For example, suppose agents must decide whether to enter a new environment, such as a school, and they care about the types of people they expect to encounter. This situation invariably leads to negative interaction effects, because preferring one group to be in the majority is equivalent to preferring another group to be in the minority. These preferences can be analyzed by using the same techniques outlined in Section 2 of the paper.

Consider a model with two social groups: a and b .² Let λ_a denote the share of people in group a , and let $s_a(\omega_i)$ denote the probability of being in group a given that one chooses ω_i . The payoff from the choice ω_i depends on the composition of agents who make that choice.

$$U_i(\omega_i|k) = v_k(\omega_i) + J_k s_a(\omega_i) + \epsilon_i(\omega_i), \quad \text{for } k \in \{a, b\}.$$

Under this framework, the parameter J_k indicates how much people in group k benefit from associating with members of group a . Both $v_k(\cdot)$ and $\epsilon_i(\cdot)$ are specified exactly as before.

In equilibrium, the expected composition of agents making a choice must be consistent with individually optimal decisionmaking. By Bayes' rule, every equilibrium must satisfy:

$$s_a(\omega_i) = \frac{\lambda_a P(\omega_i|a)}{\lambda_a P(\omega_i|a) + (1 - \lambda_a) P(\omega_i|b)}, \quad \text{for } \omega_i \in \{0, 1\},$$

where $P(\omega_i = 1|k) = F_{\varepsilon|k}(h_k + J_k(s_a(1) - s_a(0)))$. By Brouwer's FPT, a solution exists.

Uniqueness and dynamic stability of equilibria depend on the Jacobian matrix $\mathbf{D} \in \mathbb{R}^{2 \times 2}$ of the system above. Letting $\beta_k = f_{\varepsilon|k}(h_k + J_k(s_a(1) - s_a(0)))$, I can write this matrix as:

$$\mathbf{D} = \begin{bmatrix} D_{11} & -D_{11} \\ -D_{22} & D_{22} \end{bmatrix}, \quad \text{where: } \begin{cases} D_{11} = s_a(0)[1 - s_a(0)] \left(\frac{J_a \beta_a}{P(0|a)} - \frac{J_b \beta_b}{P(0|b)} \right) \\ D_{22} = s_a(1)[1 - s_a(1)] \left(\frac{J_a \beta_a}{P(1|a)} - \frac{J_b \beta_b}{P(1|b)} \right) \end{cases}$$

In this model, the amount of diversity in the network directly shapes how strongly network composition responds to changes in group behavior. Specifically, each row of the Jacobian matrix $\mathbf{D}$ is scaled by the product $s_a(\omega_i)[1 - s_a(\omega_i)]$, which is largest when the two groups

²I focus on the two-group case for simplicity, but this framework may be generalized to cases with many groups.

are equally represented, and approaches zero as the network becomes homogeneous. Thus, holding all other factors fixed, a change in one group's probability of choosing the action $\omega_i = 1$ will have a larger effect on network composition when the network is more diverse.

Suppose that all agents prefer to associate with their own group, so $J_a \geq 0$ and $J_b \leq 0$. In this case, the matrix $\mathbf{D}$ satisfies Assumption A. Hence, there is almost always a locally stable equilibrium—even if the spillover effects are strong enough to generate multiplicity.

If agents prefer to associate with members of the other group, so $J_a < 0$ or $J_b > 0$, then Assumption A may fail, and the model may not have any locally stable equilibria. Consider, for example, a model of school choice in which all households prefer schools with higher-achieving peers. If these preferences are sufficiently weak, then a unique, stable equilibrium exists. However, if households of students with lower prior achievement respond strongly to peer composition, a locally stable equilibrium may not exist: small shifts toward schools with higher-achieving peers can alter their composition enough to trigger larger shifts back.

APPENDIX B: DETAILS ON IDENTIFICATION AND ESTIMATION

B.1. Identification with Individual-Level Covariates

Suppose that the outcome equation is modified to allow for exogenous covariates $X_i \in \mathbb{R}^r$. Let $\omega_i = \mathbb{1}\{X_i'c + h_k + \alpha_n + \sum_{\ell=1}^K J_{k\ell}m_n^\ell + \varepsilon_i \geq 0\}$ where $P(\varepsilon_i \leq z|X_i, k, \alpha_n) = F_{\varepsilon|k}(z)$ and $P(X_i \leq x|k, \alpha_n) = F_{X|k}(x)$. Then $m_n^k = \int E(\omega_i|X_i, k, \alpha_n, \{m_n^\ell\}_{\ell=1}^K) dF_{X|k}$, where:

$$E(\omega_i|X_i, k, \alpha_n, \{m_n^\ell\}_{\ell=1}^K) = F_{\varepsilon|k}\left(h_k + \alpha_n + X_i'c + \sum_{\ell=1}^K J_{k\ell}m_n^\ell\right), \quad \text{for } k = 1, \dots, K.$$

In the presence of these covariates, Lemma 2 must be adapted. I do so in the following way.

LEMMA 2—Sufficiency Claim: *For any agents i and j in groups k_1 and k_2 , respectively, and network n : $E(\omega_i|X_i, k_1, \alpha_n, \{m_n^\ell\}_\ell) = E(\omega_i|X_i, X_j, k_1, \{m_n^\ell\}_\ell, E(\omega_j|X_j, k_2, \alpha_n, \{m_n^\ell\}_\ell))$.*

PROOF: As $F_{\varepsilon|k_2}$ is strictly increasing, its inverse exists. By this property, I can write:

$$\alpha_n = F_{\varepsilon|k_2}^{-1}\left(E(\omega_j|X_j, k_2, \alpha_n, \{m_n^\ell\}_\ell)\right) - h_{k_2} - X_j'c - \sum_{\ell=1}^K J_{k_2\ell}m_n^\ell.$$

By plugging this expression for α_n into the definition of $E(\omega_i|X_i, k_1, \alpha_n, \{m_n^\ell\}_\ell)$, I obtain:

$$\begin{aligned} E(\omega_i|X_i, k_1, \alpha_n, \{m_n^\ell\}_\ell) &= F_{\varepsilon|k_1}\left(h_{k_1} - h_{k_2} + (X_i - X_j)'c \right. \\ &\quad \left. + \sum_{\ell=1}^K (J_{k_1\ell} - J_{k_2\ell})m_n^\ell + F_{\varepsilon|k_2}^{-1}\left(E(\omega_j|X_j, k_2, \alpha_n, \{m_n^\ell\}_\ell)\right)\right). \end{aligned}$$

Q.E.D.

To prove Theorem 1 (semiparametric identification) with covariates X_i , I require that there is some k such that $\text{supp}(X|k)$ is not contained in a proper affine subspace of $\mathbb{R}^r$. Note that this condition is weaker than Assumption B.3. Yet, it is sufficient to show that c , $h_{k_1} - h_{k_2}$ and $\{J_{k_1\ell} - J_{k_2\ell}\}_\ell$ are identified for all k_1 and k_2 . Consider the following result and proof.³

THEOREM 1: *Suppose Assumptions B.1 and B.2 hold, and assume m_n^k is observed for all networks n and agent types k . Also, suppose that there is some type k where $\text{supp}(X|k)$ is not contained in a proper affine subspace of $\mathbb{R}^r$. If the functions $\{F_{\varepsilon|k}\}_k$ are known, then:*

- (i) *With no added assumptions, c , $\{h_{k_1} - h_{k_2}\}_{k_1, k_2}$ and $\{J_{k_1\ell} - J_{k_2\ell}\}_{k_1, k_2, \ell}$ are identified.*
- (ii) *If $\alpha_n = W_n' d$ for an observed vector W_n , then c , d , $\{h_k\}_k$, and $\{J_{k\ell}\}_{k, \ell}$ are identified.*

PROOF: Fix some network n and define the term $\zeta_n^k = h_k + \alpha_n + \sum_{\ell=1}^K J_{k\ell} m_n^\ell$. I write:

$$E(\omega_i | X_i, k, \alpha_n, \{m_n^\ell\}_{\ell=1}^K) = F_{\varepsilon|k}(\zeta_n^k + X_i' c)$$

To recover c , I must show that $F_{\varepsilon|k}(\zeta_n^k + X_i' c) = F_{\varepsilon|k}(\hat{\zeta}_n^k + X_i' \hat{c})$ implies $c = \hat{c}$. Since $F_{\varepsilon|k}$ is known, this property holds by Proposition 5 of [Manski \(1988\)](#). It follows that c is identified.

Having demonstrated that c can be recovered, I now focus on the other parameters. Consider any two social groups k_1 and k_2 , and then define the function $\phi : \mathbb{R} \rightarrow [0, 1]$ such that:

$$\phi(\nu) = \int F_{\varepsilon|k_1}((X_i - X_j)' c + \nu) dF_{X|k_1},$$

where X_j is chosen from $\text{supp}(X|k_2)$ and the integral is evaluated over the conditional support of X_i given k_1 . By definition, $\phi(\cdot)$ is nonlinear and monotonically increasing in ν .

For any n , let $\nu_n = h_{k_1} - h_{k_2} + \sum_{\ell=1}^K (J_{k_1\ell} - J_{k_2\ell}) m_n^\ell + F_{\varepsilon|k_2}^{-1}(E(\omega_j | X_j, k_2, \alpha_n, \{m_n^\ell\}_{\ell=1}^K))$. By Lemma 2, the expected average choice $m_n^{k_1}$ equals $\phi(\nu_n)$. Because $\phi(\cdot)$ is monotonic:

$$\begin{aligned} m_n^{k_1} &= \int F_{\varepsilon|k_1} \left(h_{k_1} - h_{k_2} + (X_i - X_j)' c + \sum_{\ell=1}^K (J_{k_1\ell} - J_{k_2\ell}) m_n^\ell + F_{\varepsilon|k_2}^{-1}(E(\omega_j | X_j, k_2, \alpha_n, \{m_n^\ell\}_{\ell=1}^K)) \right) dF_{X|k_1} \\ &= \int F_{\varepsilon|k_1} \left(\overline{h_{k_1} - h_{k_2}} + (X_i - X_j)' c + \sum_{\ell=1}^K \overline{(J_{k_1\ell} - J_{k_2\ell})} m_n^\ell + F_{\varepsilon|k_2}^{-1}(E(\omega_j | X_j, k_2, \alpha_n, \{m_n^\ell\}_{\ell=1}^K)) \right) dF_{X|k_1} \end{aligned}$$

is satisfied if and only if $\sum_{\ell=1}^K [\overline{(J_{k_1\ell} - J_{k_2\ell})} - (J_{k_1\ell} - J_{k_2\ell})] m_n^\ell = (h_{k_1} - h_{k_2}) - \overline{(h_{k_1} - h_{k_2})}$ for all networks in the population. By the rank condition in Assumption B.2, it follows that:

$$h_{k_1} - h_{k_2} = \overline{h_{k_1} - h_{k_2}} \quad \text{and} \quad J_{k_1\ell} - J_{k_2\ell} = \overline{J_{k_1\ell} - J_{k_2\ell}},$$

³In these results and the main text, whenever network-level covariates W_n are used, Assumption B.2 is understood to require $\text{rank}(\text{Var}[(W_n', m_n')']) = \dim(W_n) + K$, so the covariance matrix of $(W_n', m_n')'$ has full rank.

for all $\ell \in \{1, \dots, K\}$. As k_1 and k_2 are chosen arbitrarily, this holds for all type pairs. The proof in the case where $\alpha_n = W'_n d$, corresponding to part (ii) of the theorem, is analogous.

Q.E.D.

For Theorem 2, no modifications to the proof are needed, since the result already incorporates individual-level covariates. Theorem 3 also extends to finitely supported covariates. Let $Y_n^{k,x}$ be the corrected outcome among agents with type k and covariates $X_i = x$, satisfying $E(Y_n^{k,x} | \alpha_n, m_n) = h_k + \alpha_n + x'c + \sum_{\ell} J_{k\ell} m_n^{\ell}$. Comparing two covariate values in the same type and network gives $E(Y_n^{k,x} - Y_n^{k,x_0}) = (x - x_0)'c$. The affine-support condition above identifies c , since the differences $x - x_0$ span $\mathbb{R}^r$ for at least one type. Once c is identified, define $G_k(t; c) = \int F_{\varepsilon|k}(t + x'c) dF_{X|k}(x)$. The IV construction uses $G_k^{-1}(\cdot; c)$ in place of $F_{\varepsilon|k}^{-1}$, subject to the corresponding regularity and bias-correction conditions for this inverse.

B.2. Additional Details on the Measurement Error Correction

This appendix subsection provides an example of a sequence of partitioning rules for which $\mathbf{E}_{k,P}$ is nonsingular at each P and $E(Y_n^k | m_n, W_n) = q_k(m_n^k)$. Consider a sampling rule that, for any network and type, selects a subset a independently of (m_n, W_n) , leaving b nonempty. Suppose that a has sizes in $\{2, 4, 8, \dots\}$, with positive probability on each size.⁴ Let a' be a random half of a . Set $\kappa(1, n) = a$ and, for $j \geq 1$, $\kappa(j+1, n) = a'$ if $I_{n,a}^k = 2^{j+1}$ and a otherwise.

To verify that this sequence of partitioning rules delivers identification as $P \rightarrow \infty$, first note that the matrix $\mathbf{E}_{k,P}$ is nonsingular at any P , since the partitioning rules create linearly independent changes in the inverse-size moments by halving subgroups of distinct sizes. As P grows, the limiting weight whenever $I_{n,a}^k = 2^{j+1}$ is equal to $\Omega_n^k = H_j / \pi_j$, where I define $H_j = P(I_{n,a}^k \leq 2^{j+1})$ and $\pi_j = P(I_{n,a}^k = 2^{j+1})$. To see why, observe that if $j = 0$, every rule selects a , so $\Omega_n^k = 1 = H_0 / \pi_0$. Next, if $j \geq 1$, then letting e_1 denote $(1, 0, \dots, 0)'$, I obtain:

$$\begin{aligned} \Omega_n^k &= \lim_{P \rightarrow \infty} \sum_{p=1}^P [\mathbf{E}_{k,P}^{-1} e_1]_p \mathbb{1}\{\kappa(p, n) = a\} \\ &= \lim_{P \rightarrow \infty} \left[\sum_{p=1}^P [\mathbf{E}_{k,P}^{-1} e_1]_p - [\mathbf{E}_{k,P}^{-1} e_1]_{j+1} \right] \\ &= \lim_{P \rightarrow \infty} \left(1 - [\mathbf{E}_{k,P}^{-1} e_1]_{j+1} \right) \\ &= \lim_{P \rightarrow \infty} \left[1 + \frac{H_{j-1}}{\pi_j} + O\left(\frac{2^{-(P-j)(P-j+1)/2}}{\pi_j} \right) \right] = 1 + \frac{H_{j-1}}{\pi_j} = \frac{H_j}{\pi_j}, \end{aligned}$$

⁴The network size can depend on (m_n, W_n) . The independence condition applies only to the selected size of a subgroup a . Additionally, in this example, the probabilities of larger subgroup sizes must not decline too quickly: setting $\liminf_{j \rightarrow \infty} P(I_{n,a}^k = 2^{j+1}) / P(I_{n,a}^k = 2^j) > 1/2$ is sufficient for the outcome Y_n^k to have a finite variance.

where the fourth equality follows from Lagrange interpolation. By construction, the weights depend only on the specified subgroup sizes $I_{n,a}^k$, not on the underlying model parameters.

I next verify that $E(Y_n^k | m_n, W_n) = q_k(m_n^k)$. For this step, take q_k and its derivative to be bounded; treatment of endpoints is discussed below. Denote $E[q_k(\bar{\omega}_{n,a}^k) | m_n^k = m, I_{n,a}^k = 2^{j+1}]$ by $\mu_j(m)$. Then $\mu_j(m) \rightarrow q_k(m)$ as $j \rightarrow \infty$. Since a' is a random half of a , when $I_{n,a}^k = 2^{j+1}$ with $j \geq 1$ its corresponding expectation is $\mu_{j-1}(m)$. Using the identity $\Omega_n^k = H_j / \pi_j$ gives:

$$\begin{aligned} E(Y_n^k | m_n, W_n) &= \sum_{j=0}^{\infty} \pi_j E \left[\Omega_n^k q_k(\bar{\omega}_{n,a}^k) + (1 - \Omega_n^k) q_k(\bar{\omega}_{n,a'}^k) \mid m_n, W_n, I_{n,a}^k = 2^{j+1} \right] \\ &= H_0 \mu_0(m_n^k) + \sum_{j=1}^{\infty} \pi_j \left[\frac{H_j}{\pi_j} \mu_j(m_n^k) + \left(1 - \frac{H_j}{\pi_j} \right) \mu_{j-1}(m_n^k) \right] \\ &= \lim_{R \rightarrow \infty} \left[H_0 \mu_0(m_n^k) + \sum_{j=1}^R \left(H_j \mu_j(m_n^k) - H_{j-1} \mu_{j-1}(m_n^k) \right) \right] \\ &= \lim_{R \rightarrow \infty} H_R \mu_R(m_n^k) = q_k(m_n^k). \end{aligned}$$

The sum in the third line telescopes. As $H_R \rightarrow 1$, it follows that $E(Y_n^k | m_n, W_n) = q_k(m_n^k)$.

Endpoint convention. A valid convention, when the equilibrium choice probabilities are uniformly bounded away from zero and one, is to choose $\delta \in (0, 1/2)$ so that $m_n^k \in [\delta, 1 - \delta]$ almost surely, and evaluate empirical inverse responses using a bounded, smooth function $\tilde{q}_k$ on $[0, 1]$ that agrees with q_k on $[\delta, 1 - \delta]$. This function would have a bounded derivative and preserve all derivatives at the true probabilities. Hence, the finite-order bias expansion remains valid, and the telescoping argument still delivers $E(Y_n^k | m_n, W_n) = q_k(m_n^k)$ exactly.

Network sizes and practical implementation. Exact identification under this construction uses $P \rightarrow \infty$ and therefore requires subgroup sizes, and hence the available network sizes, to have a large support. Importantly, however, every network in the data can remain finite: consistency only requires that the number of networks grow under a fixed sampling scheme, for a fixed distribution of network sizes. Moreover, in practice, very few partitioning rules can suffice for precise estimation. For example, Table 2 in the main text shows that $P = 1$ already yields accurate estimates over a variety of specifications with fixed finite networks.

B.3. Asymptotic Properties of the IV Estimator

Consider an i.i.d. sample of networks, with outcomes Y_n^k as defined in (16). Maintain Assumptions B.1 and B.2. Additionally, assume that $Z_n Y_n^k$ and $Z_n X_n'$ have finite second

moments for all k . First, consider the estimand β_k defined in Theorem 3(ii). Its estimator is:

$$\hat{\beta}_k = \left(\frac{1}{N} \sum_{n=1}^N Z_n X'_n \right)^{-1} \left(\frac{1}{N} \sum_{n=1}^N Z_n Y_n^k \right).$$

Note that $\hat{\beta}_k \xrightarrow{p} \beta_k$ as the number of networks N grows large.⁵ Additionally, it follows that:

$$\sqrt{N}(\hat{\beta}_k - \beta_k) = \left(\frac{1}{N} \sum_{n=1}^N Z_n X'_n \right)^{-1} \left(\frac{1}{\sqrt{N}} \sum_{n=1}^N Z_n [Y_n^k - X'_n \beta_k] \right) \xrightarrow{d} \mathcal{N}(0, Q\Omega Q'),$$

where $Q = E(Z_n X'_n)^{-1}$ and $\Omega = \text{Var}(Z_n [Y_n^k - X'_n \beta_k])$. By the sample analogue principle, I can construct a consistent estimator for the asymptotic variance of $\hat{\beta}_k$ from $\hat{Q}\hat{\Omega}\hat{Q}'$, where:

$$\hat{Q} = \left(\frac{1}{N} \sum_{n=1}^N Z_n X'_n \right)^{-1} \quad \text{and} \quad \hat{\Omega} = \frac{1}{N} \sum_{n=1}^N Z_n [Y_n^k - X'_n \hat{\beta}_k]^2 Z'_n.$$

Next, consider the estimator $\hat{\beta}_{k_1, k_2}$, corresponding to the estimand β_{k_1, k_2} in Theorem 3(i):

$$\hat{\beta}_{k_1, k_2} = \left(\frac{1}{N} \sum_{n=1}^N Z_n X'_n \right)^{-1} \left(\frac{1}{N} \sum_{n=1}^N Z_n [Y_n^{k_1} - Y_n^{k_2}] \right).$$

As in the previous case, $\hat{\beta}_{k_1, k_2} \xrightarrow{p} \beta_{k_1, k_2}$ as N grows large. Moreover, the estimator satisfies:

$$\sqrt{N}(\hat{\beta}_{k_1, k_2} - \beta_{k_1, k_2}) \xrightarrow{d} \mathcal{N}(0, Q\Omega Q'),$$

where $Q = E(Z_n X'_n)^{-1}$ and $\Omega = \text{Var}(Z_n [Y_n^{k_1} - Y_n^{k_2} - X'_n \beta_{k_1, k_2}])$. As before, by the sample analogue principle, I can construct a consistent estimator $\hat{Q}\hat{\Omega}\hat{Q}'$ for $Q\Omega Q'$, where I define:

$$\hat{Q} = \left(\frac{1}{N} \sum_{n=1}^N Z_n X'_n \right)^{-1} \quad \text{and} \quad \hat{\Omega} = \frac{1}{N} \sum_{n=1}^N Z_n [Y_n^{k_1} - Y_n^{k_2} - X'_n \hat{\beta}_{k_1, k_2}]^2 Z'_n.$$

APPENDIX C: SUPPLEMENT TO THE MONTE CARLO SIMULATIONS

C.1. Details on the Simulation Design

For the first set of Monte Carlo simulations (Table 2, Specification A), choices are simulated from a data generating process (DGP) with $K = 1$ and $\alpha_n = W'_n d$. Outcomes are given by:

$$\omega_i = \mathbb{1} \left\{ h + W'_n d + Jm_n + \varepsilon_i \geq 0 \right\}, \quad \text{for } n = 1, \dots, N,$$

⁵Specifically, consistency is satisfied because $\frac{1}{N} \sum_{n=1}^N Z_n X'_n \xrightarrow{p} E(Z_n X'_n)$ and $\frac{1}{N} \sum_{n=1}^N Z_n Y_n^k \xrightarrow{p} E(Z_n Y_n^k)$ by the (weak) law of large numbers. Consequently, it follows from the continuous mapping theorem that $\hat{\beta}_k \xrightarrow{p} \beta_k$.

where $m_n = F_{\varepsilon|k}(h + W_n' d + J m_n)$ and W_n takes equally spaced values on the interval $[-2, 2]$. I assign the parameter values: $J = 0.5$, $h = 1$, and $d = 1$. I also assume $\varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \text{Logistic}(0, 1)$. This same design is used to generate Figure 1, which plots the bias of the OLS estimand.⁶

For the second set of Monte Carlo simulations (Table 2, Specification B), choices are simulated from a DGP with $K = 2$ and α_n treated as unobservable. Outcomes are given by:

$$\omega_i = \mathbb{1} \left\{ h_k + \alpha_n + J_{k1} m_n^1 + J_{k2} m_n^2 + \varepsilon_i \geq 0 \right\}, \quad \text{for } n = 1, \dots, N,$$

where $m_n^k = F_{\varepsilon|k}(h_k + \alpha_n + J_{k1} m_n^1 + J_{k2} m_n^2)$ for $k \in \{1, 2\}$, and α_n takes equally spaced values on the interval $[-2, 2]$. I set $h_1 = 1$ and $h_2 = 0$, and I define $\mathbf{J}$ such that: $J_{11} = J_{22} = 1.5$ and $J_{12} = J_{21} = -1$. As in the first DGP, I assume that $\varepsilon_i|k \stackrel{\text{i.i.d.}}{\sim} \text{Logistic}(0, 1)$ for $k \in \{1, 2\}$. For all cases with $K = 2$, each network contains equal numbers of agents of the two types.

For each network, I compute the equilibrium choice probabilities by fixed-point iteration on (9), then simulate individual choices conditional on these probabilities. Under both specifications, the equilibrium mapping is a contraction: the standard logistic density is bounded by $1/4$, and the maximum absolute row sum of $\mathbf{J}$ is less than four. Thus, each network has a unique equilibrium, and fixed-point iteration converges from any initial probability vector. Note that the contraction property is not required for the validity of the estimation; yet, it is useful here for guaranteeing convergence of the numerical algorithm in these simulations.

C.2. Performance with Very Small Networks

Table A.1 extends the simulation designs in Table 2 of the main text to very small networks, containing 4 to 10 agents per network-type cell. Under this specification, I vary the number of networks from 2,000 to 5,000 and compare the estimators' mean squared errors, average estimated standard errors, and 95% confidence interval coverage rates. All other aspects of the simulation design are the same as in Table 2 and as described in Appendix C.1 above.

Holding network sizes fixed, increasing the number of networks substantially reduces the IV estimator's mean squared errors and standard errors, while those of OLS remain largely unchanged. The IV estimator also achieves coverage rates close to the nominal 95% level across all designs, whereas OLS coverage is essentially zero throughout. These properties hold even in the smallest designs, where there are only 4 or 6 agents per network-type cell. The results in Table A.1 therefore illustrate that the paper's IV estimator can perform reliably with few agents per network, provided that the data contain sufficiently many networks.

⁶While the model in the example presented in Section 3.3 of the main text, given in (13), technically has choices scaled to take values in $\{-1, 1\}$, the simulations throughout adopt a logit specification where choices lie in $\{0, 1\}$.

TABLE A.1
OLS AND IV PERFORMANCE BY NETWORK-TYPE CELL SIZE AND NUMBER OF NETWORKS

Number of Networks	Agents per (n, k) Cell	Specification A ($K = 1$)						Specification B ($K = 2$)					
		$\hat{h}$		$\hat{d}$		$\hat{J}$		$\widehat{h_1 - h_2}$		$\widehat{J_{11} - J_{21}}$		$\widehat{J_{22} - J_{12}}$	
		OLS	IV	OLS	IV	OLS	IV	OLS	IV	OLS	IV	OLS	IV
N = 2,000	4	16.53	6.39	0.81	0.35	33.19	11.97	2.68	2.38	19.57	8.12	11.68	2.57
		(0.06)	(2.68)	(0.02)	(0.63)	(0.07)	(3.64)	(0.06)	(1.51)	(0.09)	(2.84)	(0.06)	(1.61)
		[0.00]	[0.92]	[0.00]	[0.93]	[0.00]	[0.91]	[0.00]	[0.98]	[0.00]	[0.98]	[0.00]	[0.96]
	6	14.18	2.94	0.72	0.16	27.77	5.42	2.57	0.66	15.47	2.19	8.16	0.69
		(0.06)	(1.75)	(0.02)	(0.41)	(0.08)	(2.38)	(0.06)	(0.77)	(0.10)	(1.45)	(0.06)	(0.82)
		[0.00]	[0.94]	[0.00]	[0.95]	[0.00]	[0.94]	[0.00]	[0.96]	[0.00]	[0.96]	[0.00]	[0.95]
	8	12.52	2.10	0.65	0.12	24.23	3.89	2.29	0.61	12.70	2.10	6.32	0.65
		(0.06)	(1.50)	(0.02)	(0.35)	(0.08)	(2.03)	(0.06)	(0.72)	(0.09)	(1.35)	(0.06)	(0.77)
		[0.00]	[0.96]	[0.00]	[0.96]	[0.00]	[0.96]	[0.00]	[0.95]	[0.00]	[0.95]	[0.00]	[0.95]
	10	11.20	1.23	0.59	0.07	21.54	2.28	2.03	0.27	10.74	0.92	5.13	0.30
		(0.06)	(1.12)	(0.02)	(0.26)	(0.08)	(1.53)	(0.06)	(0.54)	(0.09)	(1.01)	(0.06)	(0.57)
		[0.00]	[0.96]	[0.00]	[0.96]	[0.00]	[0.96]	[0.00]	[0.96]	[0.00]	[0.95]	[0.00]	[0.95]
N = 5,000	4	16.51	4.12	0.81	0.23	33.16	7.58	2.67	0.88	19.53	2.82	11.65	0.89
		(0.04)	(1.99)	(0.01)	(0.47)	(0.05)	(2.70)	(0.04)	(0.90)	(0.06)	(1.69)	(0.04)	(0.96)
		[0.00]	[0.94]	[0.00]	[0.95]	[0.00]	[0.93]	[0.00]	[0.97]	[0.00]	[0.98]	[0.00]	[0.95]
	6	14.17	1.00	0.72	0.05	27.75	1.85	2.55	0.30	15.47	0.88	8.19	0.26
		(0.04)	(1.01)	(0.01)	(0.24)	(0.05)	(1.37)	(0.04)	(0.48)	(0.06)	(0.90)	(0.04)	(0.51)
		[0.00]	[0.95]	[0.00]	[0.95]	[0.00]	[0.95]	[0.00]	[0.93]	[0.00]	[0.95]	[0.00]	[0.94]
	8	12.54	0.67	0.65	0.04	24.25	1.23	2.30	0.20	12.74	0.72	6.33	0.23
		(0.04)	(0.91)	(0.01)	(0.21)	(0.05)	(1.23)	(0.04)	(0.45)	(0.06)	(0.84)	(0.04)	(0.48)
		[0.00]	[0.96]	[0.00]	[0.96]	[0.00]	[0.96]	[0.00]	[0.96]	[0.00]	[0.96]	[0.00]	[0.95]
	10	11.23	0.40	0.59	0.02	21.59	0.73	2.03	0.11	10.74	0.39	5.11	0.12
		(0.04)	(0.68)	(0.01)	(0.16)	(0.05)	(0.92)	(0.04)	(0.34)	(0.06)	(0.64)	(0.04)	(0.36)
		[0.00]	[0.95]	[0.00]	[0.95]	[0.00]	[0.95]	[0.00]	[0.96]	[0.00]	[0.96]	[0.00]	[0.96]

Notes. This table extends the simulation design in Table 2 to networks with 4 to 10 agents per network-type cell. It reports the mean squared errors, average estimated standard errors (in parentheses), and coverage rates for 95% confidence intervals (in brackets), based on 1,000 simulation draws. All network-type cells within a given design have equal size. All other features of the simulation design are the same as in Table 2, as described in Appendix C.1.

APPENDIX D: ROBUSTNESS ANALYSES FOR PROJECT STAR

D.1. Examining Alternative Outcome Variables

Table A.2 reports IV estimates using alternative binary outcomes that indicate whether students' math and reading scores fall in the top 25% or top 75% of the statewide distribution. The estimates in this table are statistically indistinguishable from those reported in Table 3 of the main text, which use 50% as the cutoff for defining student outcomes. This similarity suggests that the findings are robust to the choice of cutoff used to define high achievement.

TABLE A.2
PROJECT STAR: IV ESTIMATES FOR ALTERNATIVE OUTCOME VARIABLES

	Math Score		Reading Score	
	Top 25%	Top 75%	Top 25%	Top 75%
$h_m - h_f$	-0.088 (0.56)	0.625 (2.69)	0.060 (0.15)	0.211 (0.47)
$J_{mm} - J_{fm}$	5.373 (2.73)	2.021 (11.26)	5.157 (0.98)	4.738 (0.76)
$J_{ff} - J_{mf}$	5.446 (2.45)	2.351 (9.51)	5.293 (1.20)	4.988 (0.79)
Observations	5,798	5,798	5,718	5,718
School FE	Yes	Yes	Yes	Yes
1st Stage $F(\bar{\omega}_n^m)$	8.74	15.53	24.53	15.50
1st Stage $F(\bar{\omega}_n^f)$	13.72	15.24	17.32	6.69

Notes. This table shows estimates of gender differences in peer effects for two alternative binary outcomes: scoring in the top 25% and scoring in the top 75%. All the other specifications are the same as in Table 3 of the main text.

D.2. Estimates for a Model with Uniform Peer Effects

Table A.3 examines how the estimates change if peer effects are restricted to be uniform, $J_{k\ell} = J$. Under this restriction, the estimates imply that girls face lower private effort costs than boys ($h_f > h_m$), although this difference is no longer statistically significant after including school fixed effects. These results differ substantially from those of the generalized model, suggesting that allowing peer effects to differ across students can materially change the conclusions about the mechanisms behind gender differences in student achievement.

TABLE A.3
PROJECT STAR: ESTIMATES FOR A MODEL WITH UNIFORM PEER EFFECTS

	Top 50% Math Score		Top 50% Reading Score	
	(1)	(2)	(1)	(2)
$h_m - h_f$	-0.363 (0.09)	-1.199 (1.73)	-0.314 (0.10)	-1.951 (1.69)
$J_{mm} - J_{fm}$	—	—	—	—
$J_{ff} - J_{mf}$	—	—	—	—
Observations	5,798	5,760	5,718	5,681
School FE	No	Yes	No	Yes

Notes. This table presents estimates of the gender fixed-effect difference, $h_m - h_f$, for a model imposing uniform peer effects, $J_{k\ell} = J$ for all $k, \ell \in \{m, f\}$. All the other specifications are the same as in Table 3 of the main text.

D.3. Verifying Random Assignment to Classrooms

Under the experimental protocols, students within a school were randomly assigned to one of three classroom types. Yet, some schools had multiple classrooms of each type, raising the possibility of nonrandom sorting across classrooms within a given type. Although [Graham \(2008\)](#) argues that such sorting is unlikely, I follow his robustness check by restricting

the sample to schools with only one classroom of each type, eliminating the possibility of within-type sorting. The first stage is weak for this restricted sample, particularly for math scores, but the estimates remain statistically indistinguishable from the full-sample results.

TABLE A.4
PROJECT STAR: IV ESTIMATES FOR SCHOOLS WITH ONE CLASSROOM OF EACH TYPE

	Top 50% Math Score		Top 50% Reading Score	
	(1)	(2)	(1)	(2)
$h_m - h_f$	0.819 (3.90)	-1.211 (9.40)	-0.089 (0.13)	-0.109 (0.49)
$J_{mm} - J_{fm}$	7.851 (16.93)	5.235 (4.18)	4.696 (0.53)	4.172 (0.76)
$J_{ff} - J_{mf}$	9.676 (25.19)	2.781 (13.27)	4.539 (0.76)	3.933 (1.32)
Observations	1,603	1,603	1,587	1,587
School FE	No	Yes	No	Yes
1st Stage $F(\bar{\omega}_n^m)$	6.18	6.41	18.99	19.63
1st Stage $F(\bar{\omega}_n^f)$	3.24	3.36	5.94	5.91

Notes. This table presents estimates of gender differences in peer effects for an estimation sample that is restricted to schools with one classroom of each type. All the other specifications are the same as in Table 3 of the main text.

D.4. Misspecification Tests

I conduct hypothesis tests to assess whether the type-specific parameter differences depend on observed classroom characteristics. Specifically, for each of $J_{ff} - J_{mf}$, $J_{mm} - J_{fm}$, and $h_m - h_f$, I evaluate the null hypothesis of no difference along each of the following criteria:

1. high poverty classrooms ($\geq 50\%$ FRPL) and low poverty classrooms ($< 50\%$ FRPL)
2. high minority classrooms ($< 75\%$ white) and low minority classrooms ($\geq 75\%$ white)
3. more experienced teachers (> 10 years) and less experienced teachers (≤ 10 years)
4. more educated teachers (graduate deg.) and less educated teachers (no graduate deg.)
5. rural classrooms (in rural district) and urban classrooms (in urban or suburban district)

The p -values from these hypothesis tests are shown in Table A.5, with math scores in Panel A and reading scores in Panel B. For both outcome variables, I find no evidence to reject the null hypothesis that each parameter difference is equal across groups. These results support the cross-group restrictions on parameter differences, which are required for identification.

D.5. Sensitivity of Estimates to the Partitioning Rule

When estimating the model, I construct internal instruments by randomly dividing students within each classroom and type into two subsamples, a and b , and using average outcomes in subsample b as instruments for the corresponding averages in subsample a . Under the model's assumptions, these averages measure the same underlying choice probabilities with conditionally independent sampling errors. However, the estimates can nevertheless vary across partitions in a finite sample. To assess sensitivity to the partitioning rule, I re-estimate

TABLE A.5
MISSPECIFICATION TESTS: PROJECT STAR

	Large Share Poverty	Large Share Minority	Teacher Has >10 Years Experience	Teacher Has Higher Degree	Rural District
<i>Panel A. Math Score Outcome</i>					
$h_m - h_f$	0.710	0.590	0.790	0.880	0.910
$J_{mm} - J_{fm}$	0.614	0.644	0.746	0.965	0.929
$J_{ff} - J_{mf}$	0.626	0.603	0.746	0.993	0.921
<i>Panel B. Reading Score Outcome</i>					
$h_m - h_f$	0.996	0.611	0.760	0.983	0.834
$J_{mm} - J_{fm}$	0.442	0.756	0.798	0.987	0.523
$J_{ff} - J_{mf}$	0.622	0.711	0.790	0.990	0.659

Notes. This table presents p -values from one-dimensional tests of whether $h_m - h_f$, $J_{ff} - J_{mf}$, and $J_{mm} - J_{fm}$ correlate with class covariates. Outcomes indicate scoring in the top 50% statewide on the corresponding exam.

the model $M = 1,000$ times, drawing a new random partition each time while holding the estimation specification fixed. I then plot histograms of the resulting parameter estimates.

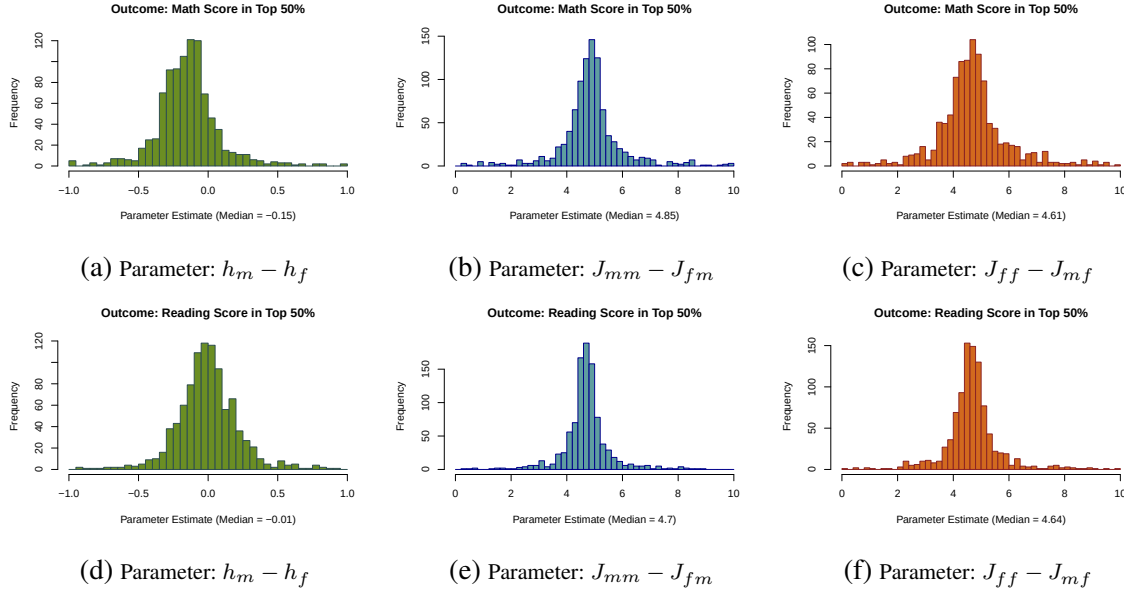
Figure A.1 reports the histograms of the parameter estimates for each outcome. The estimates appear approximately normally distributed, and variation across partitions preserves the qualitative patterns reported in Table 3. The baseline IV estimates also lie within the main mass of these distributions. Together, these results suggest that the main findings are not particularly sensitive to the random partition used to construct the internal instruments.

D.6. Finite-Sample Performance of the IV Estimator

Table A.6 examines the IV estimator's finite-sample performance in simulations that reproduce Project STAR's number of classrooms, classroom sizes, and gender composition. I calibrate the parameters to the estimates in Column (1) of Table 3, the specification without controls.⁷ I use this specification throughout all simulations and apply the same partitioning rule and standard error calculation used to obtain the estimates in Table 3. To simulate data, I follow the procedure described in Appendix C.1: running a fixed-point iteration under the logistic model, and then drawing agents' choices conditional on the resulting probabilities.

Based on 2,000 simulation draws, Table A.6 reports root mean squared errors, average estimated standard errors, and 95% confidence interval coverage rates. Panel A presents results for math scores, and Panel B presents results for reading scores. The table demonstrates that average estimated standard errors are close to the root mean squared errors, and all coverage rates are within 2.5 percentage points of the nominal 95% level. These results support the reliability of inference in finite samples that resemble the empirical application.

⁷Since only differences in the parameters are identified, I set the remaining parameters, namely the levels of h_k and $J_{k\ell}$ and the distribution of α_n , to match the observed distribution of outcomes across classrooms in the data.

FIGURE A.1.—Project STAR: Histograms of IV Estimates across $M = 1000$ Random PartitionsTABLE A.6
PROJECT STAR: FINITE-SAMPLE PERFORMANCE OF THE IV ESTIMATOR

	Target Value	Root MSE	Avg. Standard Error	95% Coverage Rate
<i>Panel A. Math Score Outcome</i>				
$h_m - h_f$	-0.112	0.366	0.355	0.956
$J_{mm} - J_{fm}$	4.229	0.355	0.359	0.968
$J_{ff} - J_{mf}$	4.127	0.363	0.354	0.956
<i>Panel B. Reading Score Outcome</i>				
$h_m - h_f$	-0.029	0.505	0.511	0.971
$J_{mm} - J_{fm}$	5.339	0.501	0.505	0.972
$J_{ff} - J_{mf}$	5.223	0.523	0.520	0.963

Notes. This table presents root mean squared errors, average estimated standard errors, and coverage rates for 95% confidence intervals, based on 2,000 simulation draws. The design exactly reproduces the number of classrooms and each classroom's size and gender composition in the Project STAR data. The simulations use the specification in Column (1) of Table 3 (no school fixed effects), with target values set to the corresponding empirical estimates.

APPENDIX E: ROBUSTNESS ANALYSES FOR PRIMARY SCHOOL DEWORMING

E.1. Differences in Treatment Uptake within Schools

Figure A.2 shows histograms of the within-school gaps in treatment uptake rates by student type. For any two types k_1 and k_2 , the size and direction of these gaps vary greatly across schools, which suggests that factors at the school level shape disparities in treatment uptake.

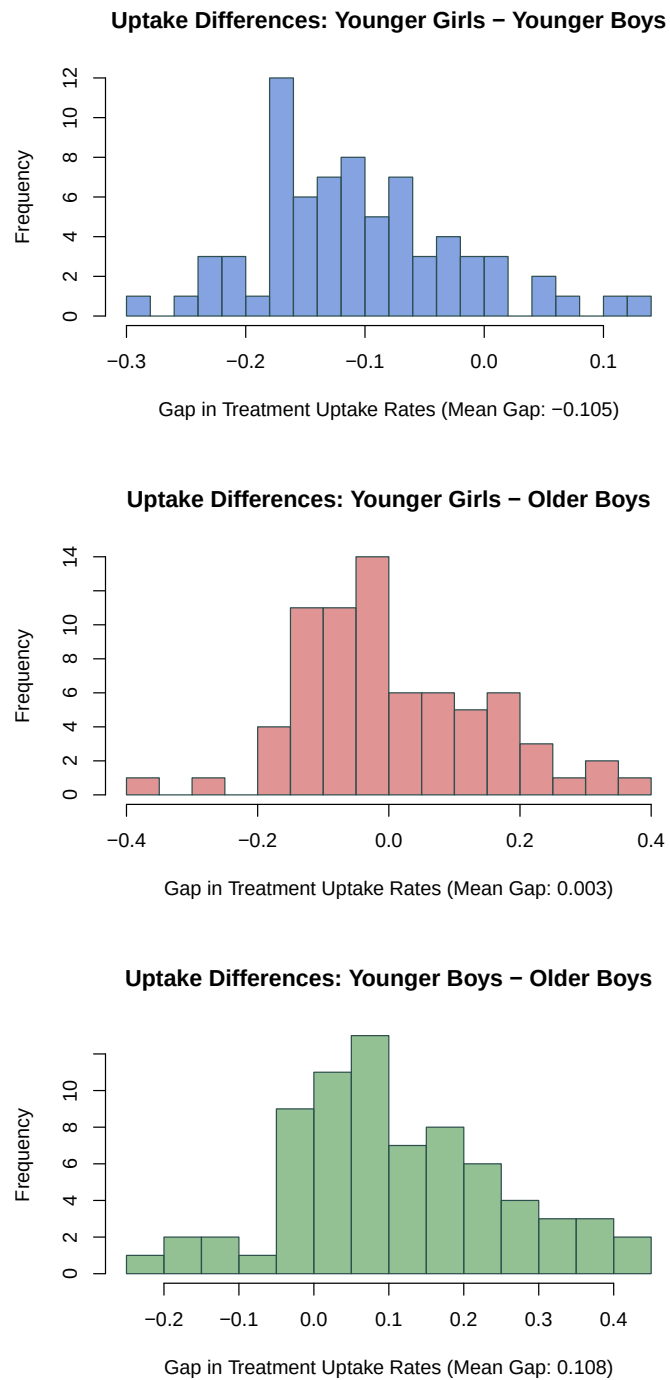


FIGURE A.2.—Within-School Differences in Treatment Uptake Rates Across Student Types

E.2. IV Estimates for First-Time Eligible Students

To better isolate contemporaneous interactions from preexisting factors that can influence treatment decisions, Table A.7 reports IV estimates for eligible students in schools receiving treatment for the first time. This sample includes Group 1 schools in 1998 and Group 2 schools in 1999. The resulting estimates are largely consistent with the full-sample results.

TABLE A.7
DEWORMING IN SCHOOLS: IV ESTIMATES FOR FIRST-TIME ELIGIBLE STUDENTS

	(1)	(2)	(3)	(4)
<i>Girls ≤ 13 and Boys ≤ 13</i>				
$h_1 - h_2$	0.338 (1.31)	0.227 (1.41)	0.779 (2.19)	1.184 (2.16)
$J_{11} - J_{21}$	10.488 (3.29)	10.777 (3.38)	10.647 (2.22)	10.973 (1.81)
$J_{12} - J_{22}$	-10.527 (2.57)	-11.105 (2.63)	-13.414 (2.77)	-14.942 (2.62)
$J_{13} - J_{23}$	-0.402 (1.21)	-0.001 (1.23)	1.369 (1.46)	1.452 (1.56)
<i>Girls ≤ 13 and Boys > 13</i>				
$h_1 - h_3$	0.494 (1.00)	0.062 (0.95)	1.079 (1.09)	2.193 (1.10)
$J_{11} - J_{31}$	5.215 (3.83)	6.338 (3.86)	5.645 (4.19)	5.226 (4.75)
$J_{12} - J_{32}$	-1.251 (3.51)	-3.497 (4.66)	-5.055 (6.07)	-5.898 (7.05)
$J_{13} - J_{33}$	-4.206 (0.72)	-2.647 (1.37)	-1.248 (1.57)	-1.086 (1.74)
<i>Boys ≤ 13 and Boys > 13</i>				
$h_2 - h_3$	0.156 (1.83)	-0.165 (1.67)	0.300 (1.71)	1.009 (2.20)
$J_{21} - J_{31}$	-5.273 (5.98)	-4.439 (5.97)	-5.003 (5.07)	-5.747 (6.08)
$J_{22} - J_{32}$	9.276 (5.14)	7.608 (5.76)	8.359 (6.28)	9.043 (7.72)
$J_{23} - J_{33}$	-3.804 (1.22)	-2.647 (1.70)	-2.617 (1.31)	-2.538 (1.31)
Year Fixed Effects	No	Yes	Yes	Yes
Student Controls	No	No	Yes	Yes
School Controls	No	No	No	Yes
1st Stage $F(\bar{\omega}_n^1)$	9.32	6.97	12.64	10.34
1st Stage $F(\bar{\omega}_n^2)$	10.02	8.05	12.08	9.74
1st Stage $F(\bar{\omega}_n^3)$	32.17	13.42	11.08	10.55

Notes. This table shows the estimated differences in spillover effects in treatment uptake for an estimation sample restricted to eligible students in schools receiving treatment for the first time: Group 1 schools in 1998 and Group 2 schools in 1999. Apart from this sample restriction, all other specifications are the same as in Table 4 of the paper.

E.3. Misspecification Tests

For each type-specific parameter difference, $h_{k_1,t} - h_{k_2,t}$ and $J_{k_1\ell} - J_{k_2\ell}$, I test the null hypothesis of equality across groups that are defined by the following school characteristics:

1. local infection rate (above vs. below median)
2. treatment group (1998 group vs. 1999 group)
3. population density within 3 km (above vs. below median)

4. population density between 3 and 6 km (above vs. below median)

Table A.8 reports the p -values from the hypothesis tests. Overall, I find very little evidence that the parameter differences vary with observable school characteristics. These results are therefore consistent with the restrictions on parameter differences underlying identification.

TABLE A.8
MISSPECIFICATION TESTS: PRIMARY SCHOOL DEWORMING

	High Local Infection Rate	Treatment Group	High Population Density (< 3km)	High Population Density (3-6km)
<i>Girls ≤ 13 and Boys ≤ 13</i>				
$h_1 - h_2$	0.954	0.914	0.053	0.748
$J_{11} - J_{21}$	0.304	0.722	0.583	0.231
$J_{12} - J_{22}$	0.120	0.772	0.381	0.122
$J_{13} - J_{23}$	0.983	0.735	0.852	0.650
<i>Girls ≤ 13 and Boys > 13</i>				
$h_1 - h_3$	0.637	0.831	0.886	0.727
$J_{11} - J_{31}$	0.158	0.806	0.277	0.281
$J_{12} - J_{32}$	0.049	0.799	0.395	0.053
$J_{13} - J_{33}$	0.348	0.754	0.746	0.715
<i>Boys ≤ 13 and Boys > 13</i>				
$h_2 - h_3$	0.343	0.751	0.367	0.999
$J_{21} - J_{31}$	0.580	0.934	0.682	0.886
$J_{22} - J_{32}$	0.866	0.866	0.993	0.709
$J_{23} - J_{33}$	0.278	0.835	0.795	0.433

Notes. This table presents p -values from one-dimensional tests of whether spillover effect differences and type fixed-effect differences correlate with observed school characteristics. The regression specification includes year fixed effects but does not add student controls. All other specifications are the same as in Table 4 of the main text.

E.4. Sensitivity of Estimates to the Partitioning Rule

As in the first empirical application, I evaluate the sensitivity of the IV estimates to the random partition used to construct the internal instruments. I re-estimate the model $M = 1,000$ times, each time drawing a new random partition, and report histograms of the resulting estimates in Figure A.3. The baseline IV estimates in Table 4 lie within the main mass of these distributions, and variation across partitions generally preserves the qualitative patterns in the baseline results. These findings suggest that the main conclusions in Section 4.2 of the main text are not especially sensitive to the partition that is used to construct instruments.

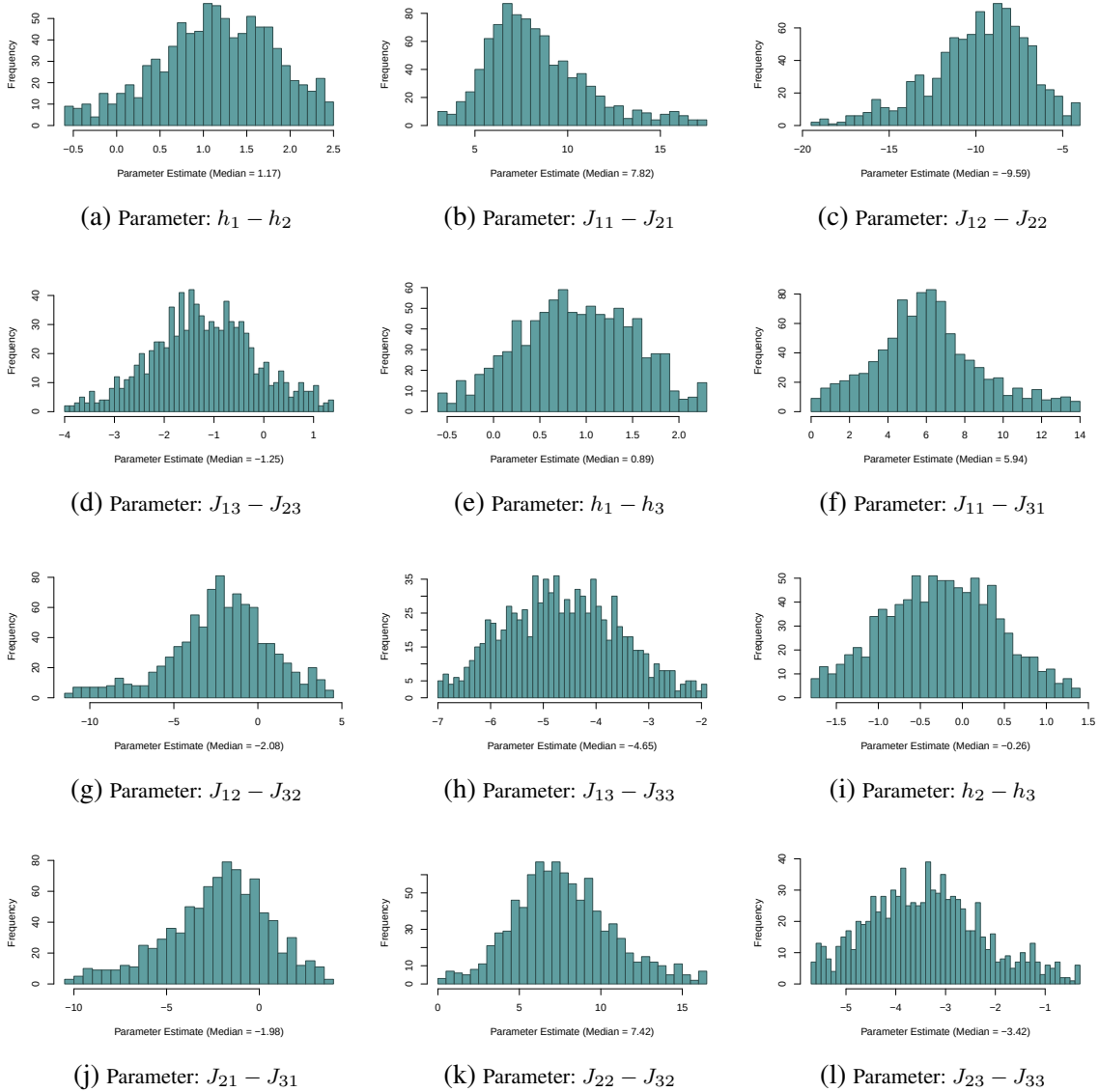


FIGURE A.3.—Primary School Deworming: Histograms of IV Estimates across $M = 1000$ Random Partitions. The estimation procedure uses the same regression specification as in Table 4, Column 4.

E.5. Finite-Sample Performance of the IV Estimator

Table A.9 assesses the IV estimator's finite-sample performance in simulations that reproduce the deworming application's number of schools, school sizes, and gender and age composition. I calibrate the parameters to estimates from the specification without year or zone fixed effects or student or school controls.⁸ For this reason, the target values differ slightly

⁸As in Appendix D.6, only differences in the parameters are identified, so I set the levels of h_k and $J_{k\ell}$, and the distribution of the network effects α_n , to match the distribution of treatment uptake across schools in the data.

from the estimates in Table 4, which includes these controls. I apply the same partitioning rule and standard error calculation used to obtain the paper’s main estimates. To simulate data, I follow the procedure described in Appendix C.1, running a fixed-point iteration under the logistic model and then drawing agents’ choices conditional on those probabilities.

Based on 2,000 simulation draws, Table A.9 reports root mean squared errors, average estimated standard errors, and coverage rates for 95% confidence intervals. The three panels present results for each pair of student types. Average estimated standard errors are close to root mean squared errors, and coverage rates range from 91% to 99%. Overall, these results support the reliability of inference in finite samples that resemble the empirical application.

TABLE A.9
PRIMARY SCHOOL DEWORMING: FINITE-SAMPLE PERFORMANCE OF THE IV ESTIMATOR

	Target Value	Root MSE	Avg. Standard Error	95% Coverage Rate
<i>Girls ≤ 13 and Boys ≤ 13</i>				
$h_1 - h_2$	0.846	0.494	0.509	0.965
$J_{11} - J_{21}$	8.069	1.210	1.147	0.965
$J_{12} - J_{22}$	-9.244	1.579	1.475	0.961
$J_{13} - J_{23}$	0.130	1.519	1.396	0.963
<i>Girls ≤ 13 and Boys > 13</i>				
$h_1 - h_3$	-0.629	0.492	0.517	0.991
$J_{11} - J_{31}$	7.563	1.304	1.176	0.929
$J_{12} - J_{32}$	-1.490	1.792	1.513	0.956
$J_{13} - J_{33}$	-5.229	1.699	1.431	0.915
<i>Boys ≤ 13 and Boys > 13</i>				
$h_2 - h_3$	-1.475	0.530	0.519	0.944
$J_{21} - J_{31}$	-0.506	1.343	1.181	0.957
$J_{22} - J_{32}$	7.754	1.721	1.521	0.932
$J_{23} - J_{33}$	-5.358	1.719	1.439	0.912

Notes. This table presents root mean squared errors, average estimated standard errors, and coverage rates for 95% confidence intervals, based on 2,000 simulation draws. The design exactly reproduces the number of schools and each school’s size and type composition in the deworming data. The simulations use the specification without year or zone fixed effects or student or school controls, with target values set to the corresponding empirical estimates.

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